Year 11 Mathematics

Shape and Space 6

10.6 Similar Shapes

Similar Shapes

STARTER · 5 MIN

Date: _____ **Title:** Similar Shapes

LEARNING OBJECTIVES

- Recognise similar shapes
- Use scale factors to find missing lengths
- Use area and volume scale factors

STARTER *Answer in your book. No calculators.*

1. Enlarge a 4 cm line by scale factor 3
2. Work out $15 \div 5$
3. Work out 3^2
4. Work out 2^3

Starter — Answers

STARTER

Tick or correct in green pen

1. $4 \times 3 = \mathbf{12\ cm}$
2. **3**
3. **9**
4. **8**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can recognise similar shapes	☐
SILVER	I can use scale factors to find missing lengths	☐
GOLD	I can use area and volume scale factors	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Similar Shapes

INTRODUCTION · 3 MIN

Similar shapes have equal angles, and their sides are in the same ratio.

Similar

One shape is an enlargement of the other.

Angles stay the same

Scale factor

k = new length ÷ old length

6 cm → 15 cm: k = 2.5

Area

Areas are multiplied by k^2.

k = 3: area × 9

Volume

Volumes are multiplied by k^3.

k = 2: volume × 8

Examples 1 and 2 — Finding Missing Lengths

EXAMPLES · 5 MIN

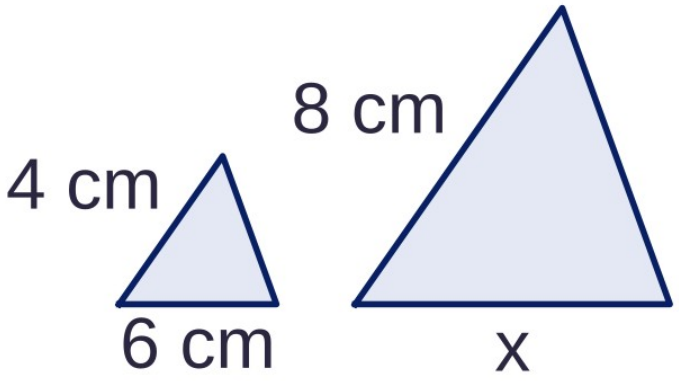
BRONZE

Watch and copy into your book

Hint scale factor = $\frac{\text{new length}}{\text{old length}}$

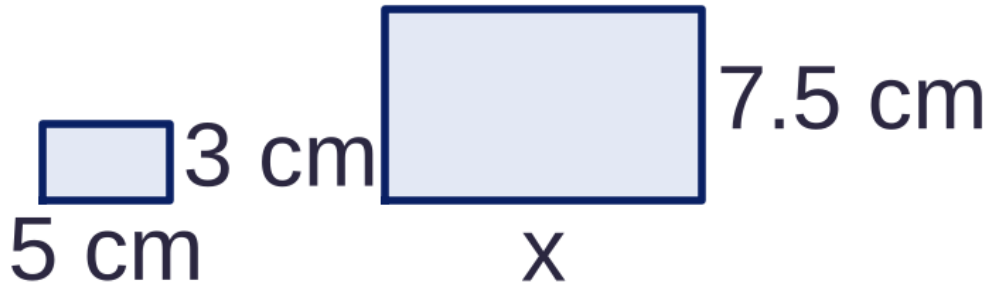
Example 1

The triangles are similar. Find x.



Example 2

The rectangles are similar. Find x.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$k = 8 \div 4$

$= 2$

$x = 6 \times 2$

$x = 12$ cm

Tip: use a pair of matching sides.

Example 2

$k = 7.5 \div 3$

$= 2.5$

$x = 5 \times 2.5$

$x = 12.5$ cm

Tip: match short side with short side.

Task 1

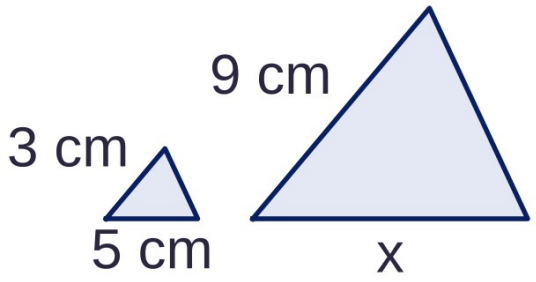
PRACTICE · 9 MIN

BRONZE

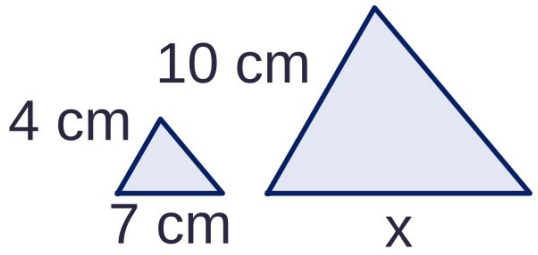
The shapes in each pair are similar. Find x.

Hint scale factor = $\frac{\text{new length}}{\text{old length}}$

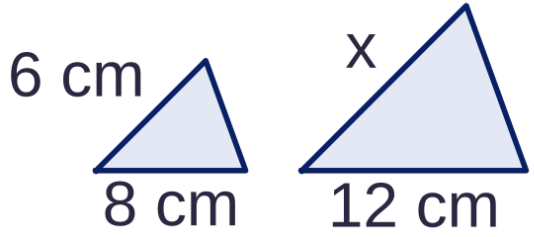
1



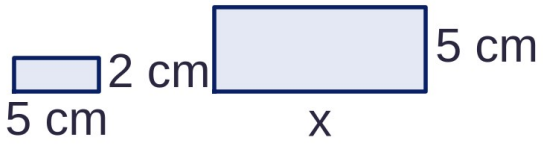
2



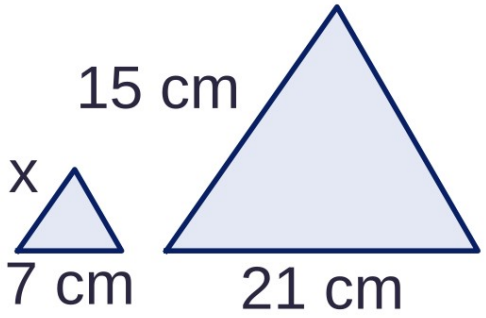
3



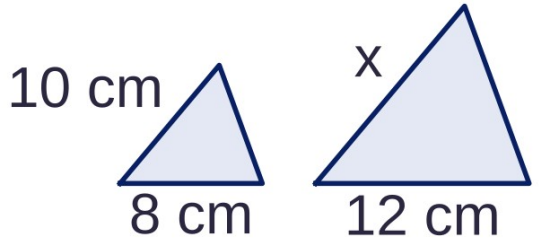
4



5



★



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $k = 3$, $x = \mathbf{15\ cm}$

2. $k = 2.5$, $x = \mathbf{17.5\ cm}$

3. $k = 1.5$, $x = \mathbf{9\ cm}$

4. $k = 2.5$, $x = \mathbf{12.5\ cm}$

5. $k = 3$, $x = \mathbf{5\ cm}$

★ $k = 1.5$, $x = \mathbf{15\ cm}$

Examples 3 and 4 — Similar Triangles

EXAMPLES · 5 MIN

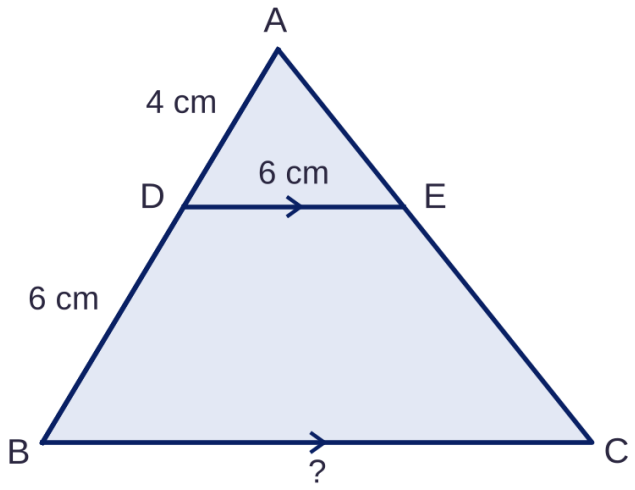
SILVER

Watch and copy into your book

Hint scale factor = $\frac{\text{new length}}{\text{old length}}$

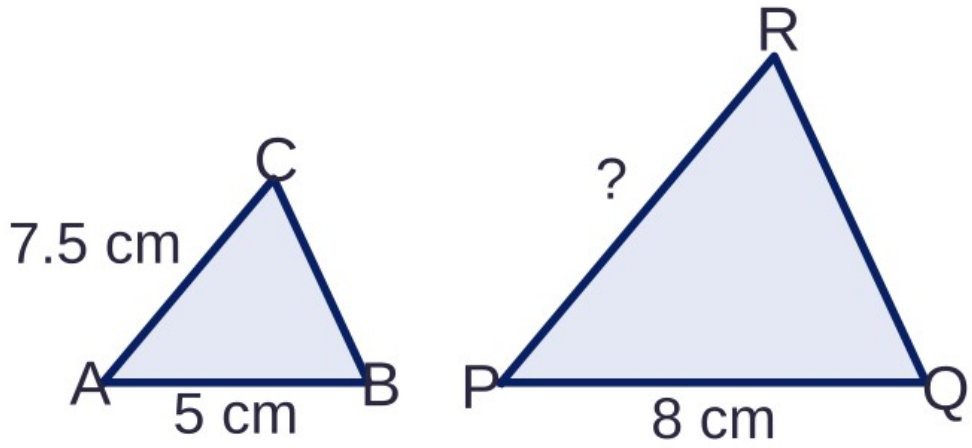
Example 3

DE is parallel to BC. Find BC.



Example 4

ABC and PQR are similar. Find PR.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

ADE and ABC are similar

$k = 10 \div 4$

$= 2.5$

$BC = 6 \times 2.5$

$= 15$ cm

Tip: parallel lines make similar triangles.

Example 4

$k = 8 \div 5$

$= 1.6$

$PR = 7.5 \times 1.6$

PR = 12 cm

Tip: AB matches PQ.

Task 2

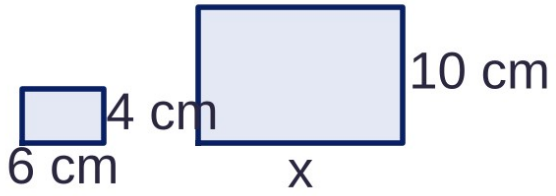
PRACTICE · 9 MIN

SILVER

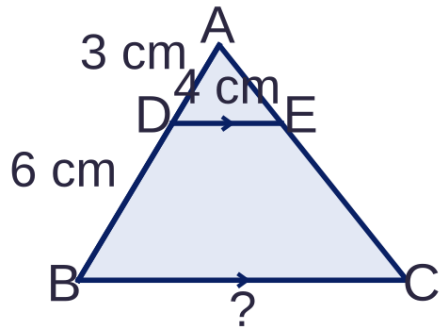
Answer in your book

Hint scale factor = $\frac{\text{new length}}{\text{old length}}$

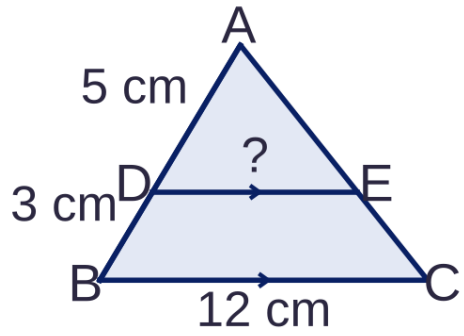
1 Similar rectangles. Find x.



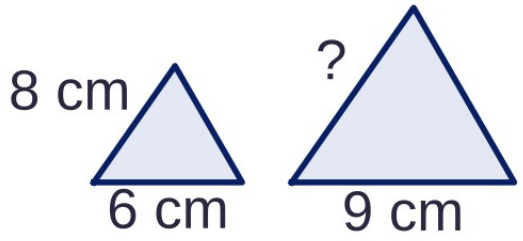
2 DE ∥ BC. Find BC.



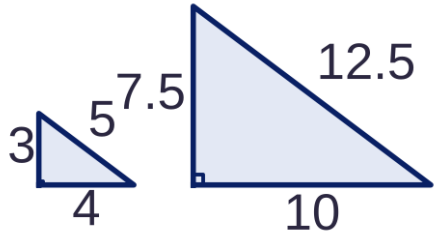
3 DE ∥ BC. Find DE.



4 Similar triangles. Find the missing side.



5 Show these triangles are similar.



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $k = 2.5$, $x =$ **15 cm**
2. $k = 3$, $BC =$ **12 cm**
3. $k = 1.6$, $DE = 12 \div 1.6 =$ **7.5 cm**
4. $k = 1.5$, $8 \times 1.5 =$ **12 cm**
5. Each ratio is 2.5, **similar**

Examples 5 and 6 — Area and Volume

EXAMPLES · 5 MIN

GOLD

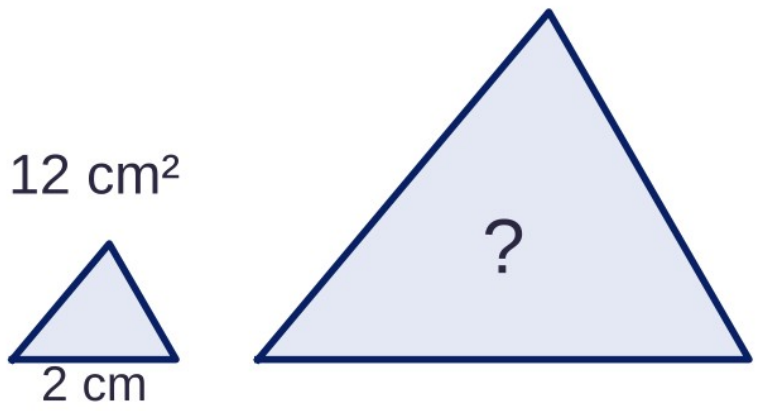
Exam-style: show all your working

Hint area factor = k^2 volume factor = k^3

Example 5

(3 marks)

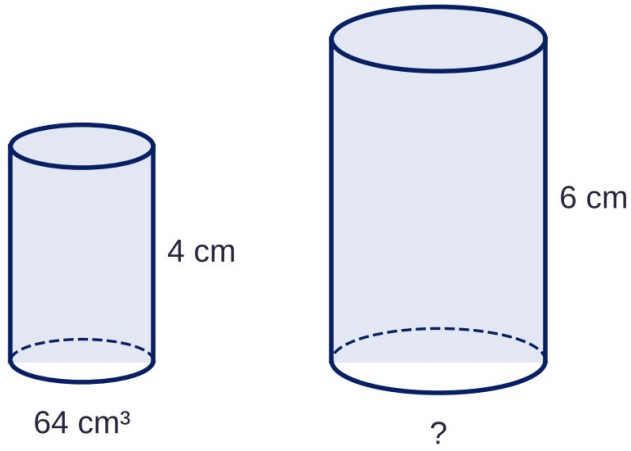
Lengths in the ratio 1 : 3. Find the large area.



Example 6

(3 marks)

Similar cylinders. Find the large volume.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$k = 3$$

$$k^2 = 9$$

$$12 \times 9 = \mathbf{108\ cm^2}$$

Tip: areas use k^2.

Example 6

$$k = 6 \div 4$$

$$= 1.5$$

$$k^3 = 3.375$$

$$64 \times 3.375 = \mathbf{216\ cm^3}$$

Tip: volumes use k^3.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

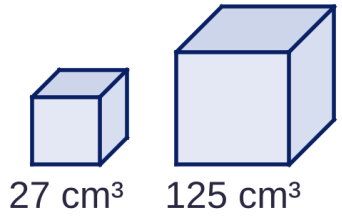
Hint area factor = k^2 volume factor = k^3

1 Lengths 2 : 5. Find the large area.



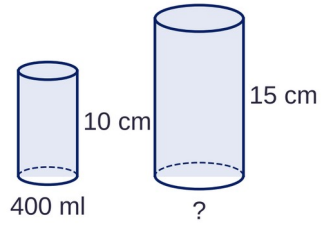
(2 marks)

2 Find the ratio of the lengths.



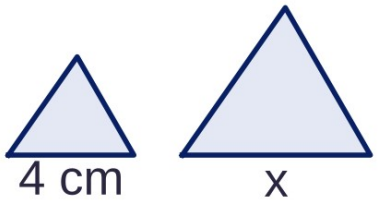
(2 marks)

3 Similar bottles. Find the large capacity.



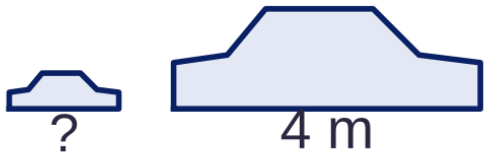
(3 marks)

4 Areas 20 cm^2 and 45 cm^2. Find x.



(3 marks)

5 Model 1 : 50. Real car is 4 m long.



(3 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $k^2 = 6.25, \quad 8 \times 6.25 = \mathbf{50\ cm^2}$

2. 3^3 and 5^3, **3 : 5**

3. $k^3 = 3.375, \quad 400 \times 3.375 = \mathbf{1350\ ml}$

4. $k^2 = 2.25, \quad k = 1.5, \mathbf{6\ cm}$

5. $400 \div 50 = 8$ cm, $\quad 2 \times 2500 = \mathbf{5000\ cm^2}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the scale factor from 4 cm to 10 cm

SILVER Lengths double. What is the area multiplied by?

GOLD Lengths are multiplied by 3. A volume of 10 cm^3 becomes?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $10 \div 4 = \mathbf{2.5}$

SILVER $2^2 = \mathbf{4}$

GOLD $3^3 \times 10 = \mathbf{270\ cm^3}$

Exam Practice — Finding Missing Lengths

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint scale factor = $\frac{\text{new length}}{\text{old length}}$

Question 1

(2 marks)

Rectangle A is 4 cm by 6 cm.

Rectangle B is similar to A.

The short side of B is 10 cm.

Find the long side of B.

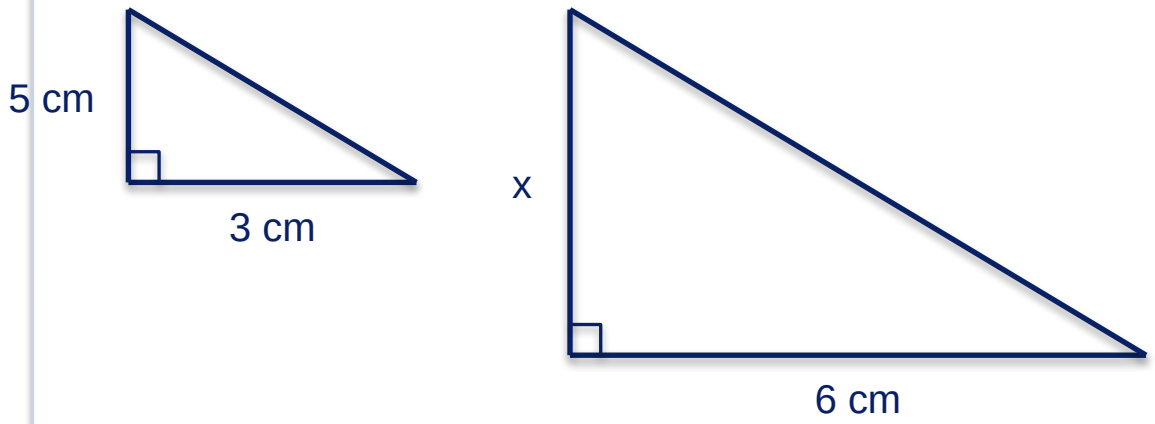
..... cm

Question 2

(2 marks)

The triangles are similar. Find x.

Diagram NOT accurately drawn



x = ...

Exam Practice — Similar Triangles

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint scale factor = $\frac{\text{new length}}{\text{old length}}$

Question 3

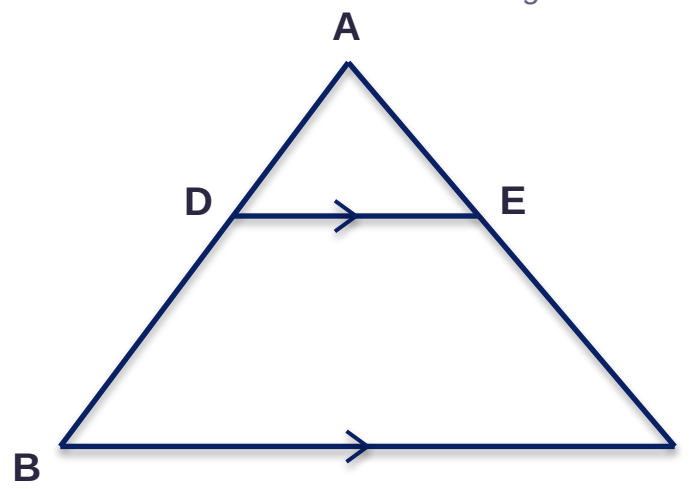
(3 marks)

DE is parallel to BC.

AD = 4 cm, AB = 10 cm, DE = 6 cm.

Find BC.

Diagram NOT accurately drawn



..... cm

Question 4

(2 marks)

Triangles PQR and XYZ are similar.

PQ = 5 cm, QR = 8 cm and XY = 7.5 cm.

Find YZ.

..... cm

Exam Practice — Area and Volume

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint area factor = k^2 volume factor = k^3

Question 5

(2 marks)

Two similar shapes have lengths in the ratio 2 : 5.

The smaller shape has an area of 12 cm^2.

Find the area of the larger shape.

..... cm^2

Question 6

(3 marks)

Two similar bottles have heights of 10 cm and 15 cm.

The smaller bottle holds 400 ml.

Find the capacity of the larger bottle.

..... ml

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

scale factor = 10 ÷ 4 = 2.5 — M1

6 × 2.5 = 15 cm — A1

Tip: match the short sides.

Question 2

scale factor = 6 ÷ 3 = 2 — M1

x = 5 × 2 = 10 cm — A1

Tip: the 3 cm and 6 cm sides match.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

scale factor = 10 ÷ 4 = 2.5 — M1

BC = 6 × 2.5 — M1

= 15 cm — A1

Tip: use AB, not DB.

Question 4

scale factor = 7.5 ÷ 5 = 1.5 — M1

YZ = 8 × 1.5 = 12 cm — A1

Tip: PQ matches XY, QR matches YZ.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

area factor = $2.5^2 = 6.25$ — M1

$12 \times 6.25 = 75\ cm^2$ — A1

Tip: square the length scale factor.

Question 6

length factor = $15 \div 10 = 1.5$ — M1

volume factor = $1.5^3 = 3.375$ — M1

$400 \times 3.375 = 1350\ ml$ — A1

Tip: capacity is a volume, so cube it.